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312 Homework 2

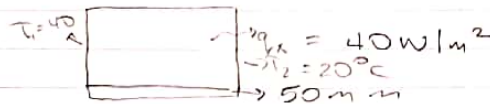
1.6 heat flux wood 50mm thick.

temp: 40°C inner 20°C outer

40 W/m²

Find thermal conductivity

Solution

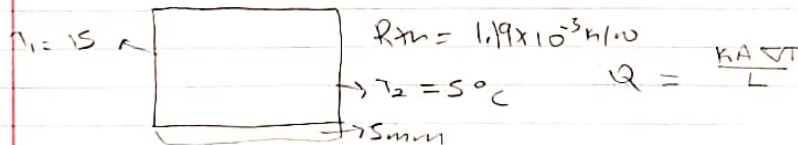


$$\text{thermal conductivity} = q_x \left(\frac{L}{T_1 - T_2} \right)$$

$$= 40 \frac{\text{W}}{\text{m}^2} \cdot \frac{0.05 \text{ m}}{313.15 \text{ K} - 293.15 \text{ K}}$$

$$k = 0.10 \text{ W/m}\cdot\text{K}$$

1.7



$$Q = \frac{\Delta T}{R_{th}}$$

$$R_{th} = \frac{L}{kA}$$

$$Q = \frac{15 - 5}{1.19 \times 10^{-3} \text{ K/W}}$$

$$= 8403.36 \text{ W} \\ = 8.403 \text{ kW}$$

$$k = \frac{0.005}{(1.19 \times 10^{-3})(1)(3)}$$

$$= 1.400 \text{ W/m}\cdot\text{K}$$

1.14



$$q_x'' = h(T_{\text{air}} - T_{\text{wall}})$$

$$= 5 \text{ W/m}^2\text{K} (20^\circ\text{C} - 16^\circ\text{C})$$

Interior air to wall = 20 W/m²

$$q_x = 20 \text{ W/m}^2\text{K} (6^\circ\text{C} - 5^\circ\text{C})$$

wall to exterior air = 20 W/m²

$$q_x = 5 \text{ W/m}^2\text{K} (16^\circ\text{C} - 20^\circ\text{C})$$

wall to interior air = -20 W/m²

Cannot tell if wall is at steady state because energy is increasing and decreasing.

1.27 Given: $400 \text{ K } T_1$, $T_2 = 97 \text{ K}$
 $L = 0.3 \text{ m}$ Emissivity = 0.25

$$a \quad E_{\text{in}} - E_{\text{out}} = 0$$

$$q_{\text{elec}} - q_{\text{rad}} = 0$$

$$q_{\text{elec}} = \epsilon_p A_p \sigma (T_1^4 - T_2^4)$$

$$q_{\text{elec}} = 0.25 (T_1 (0.3)^2 / 4) \times 5.67 \times 10^{-8} (400^4 - 97^4)$$

$$= 25.56 \text{ W}$$

$$b \quad E_{\text{in}} - E_{\text{out}} = 0$$

$$q_{\text{rad}} - m_{\text{H}_2\text{O}} h_{\text{fg}} = 0$$

$$= \frac{25.56 \text{ W}}{125 \text{ kJ/kg}} = 2.045 \times 10^{-4} \text{ kg/s}$$

$$= 0.74 \text{ kg/h}$$

$$C \quad \varepsilon = 0.09$$

$$(0.09) \left(\frac{\pi}{4}\right) (0.3^2) (5.67 \times 10^{-8}) (400^4 - 97^4)$$

$$= 9.20$$

$$\frac{9.20}{125 \times 10^3} = \underline{7.36 \times 10^{-5} \text{ kg/s}}$$

Yes